

Exam Date & Time: 11-May-2024 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MIT BLR - BTech IV Semester - End Semester Examination - April / May -2024

Computer Organization and Microprocessor Systems [IT- 2256]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain the use of following instruction.
i) RCR ii) PUSHF iii) DAS iv) IMUL v) JLS (5)
 - A)
 - B) Write an assembly language program to i) clear the screen ii) set cursor at the centre of the screen (3)
 - C) If CS=2000h, DS=3000h, SS=4000h, ES=5000h, BX=0030h
Determine the physical address for i) MOV CX, [BX] ii) ADD AX, 20[BX] (2)
- 2) Divide 11 by 3 using restoring division algorithm indicating all the steps. (5)
 - A)
 - B) Write an assembly language program to convert a 2-digit hexadecimal number available in the memory into BCD and store the result in the memory. (3)
 - C) Write a procedure to set Trap Flag (2)
- 3) Sketch the steps of Mode 2 operation of 8254. (5)
 - A)
 - B) Describe control word format of 8255 (3)
 - C) Assume there is a class of people with following grades: 69, 87, 96, 45, 75. Develop an assembly language program to find the highest grade (2)
- 4) Present Big endian and Little endian method of byte addressing with examples. (5)
 - A)
 - B) A half adder is a combinational logic circuit that has two inputs x and y. and two outputs, sum(s) and carry (c), resulting from the binary addition of x and y. Draw a half adder as a two level (3)

AND-OR circuit

- C) Given $M = 17_{10}$ and $Q = -16_{10}$, Perform multiplication using Booth's algorithm (2)
- 5) Calculate average access time experienced by a processor, if a cache hit rate .88, miss penalty is 0.015 milliseconds and cache access time is 10 microseconds (5)
- A)
- B) Consider a memory consisting of 64K words of 8 bits each. Sketch the organization to implement this memory using 16K X 1 static memory chips. (3)
- C) Define the term cycle stealing and burst mode with respect to DMA (2)

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